



11864 - Into the red: characterizing the mid-infrared emission of the circumplanetary disk around PDS70 c

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	2	PDS70- Roll1	MIRI Imaging	(2) PDS70
	3	PDS70 - Roll2	MIRI Imaging	(2) PDS70
	1	Reference F560W	MIRI Imaging	(3) Ref_for_F560W
	4	Reference F770W	MIRI Imaging	(4) Ref_for_F770W

ABSTRACT

Several challenges have so far prevented the investigation of the mid-infrared thermal emission of the circumplanetary disk (CPD) around PDS70c: angular resolution, thermal background and contrast. Here we propose to overcome all of them by combining for the first time the capabilities of the MIRI Imager with Kernel Phase Imaging (KPI), a state-of-the-art technique able to achieve 0.5 λ/D angular resolution. Our observations aim at

JWST Proposal 11864 (Created: Friday, March 13, 2026, 5:03:34PM Eastern Standard Time) - Overview

detecting and measuring the thermal emission from the CPD in F560W and F770W (5.6 and 7.7 μm), for the first time pushing its SED redwards of 5 μm . At these wavelengths the CPD flux is dominating over the atmosphere's, providing tight constraints on its temperature and emitting surface, which can be used to constrain the size of the inner cavity and the accretion mechanism. Coupled with existing and upcoming ALMA observations, these data will enable the first radiative transfer calculations of a CPD.

Demonstrating the synergy between MIRI and KPI is essential in order to enable the exploration of the innermost regions of planetary systems at moderate contrasts, opening a new observing window into a parameter space neither coronagraphy nor normal imaging can access.

OBSERVING DESCRIPTION

This program aims at detecting PDS70c with the MIRI imager using Kernel Phase Interferometry. To minimize the exposure time of single groups, we will use the SUB64 array. Mitigation of bad pixels and detector effects will occur via 4 points dither. We will observe in F560W and F770W. The number of groups has been maximized to avoid saturation based on the Cycle 1 MRS spectrum. Then, we integrated long enough to collect 10^9 photons from the system, necessary to detect PDS70c. To remove PSF we will observe at 2 roll angles and we will observe 2 reference stars of very similar brightness.

Proposal 11864 - Targets - Into the red: characterizing the mid-infrared emission of the circumplanetary disk around PDS70 c

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(2)	PDS70	RA: 14 08 10.1546 (212.0423108d) Dec: -41 23 52.57 (-41.39794d) Equinox: J2000	Proper Motion RA: -29.69699999999996 mas/yr Proper Motion Dec: -24.04100005151122 mas/yr Parallax: 0.008897500000000001" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[K stars, Protoplanetary disks]				
	(3)	Ref_for_F560W	RA: 14 07 25.5626 (211.8565108d) Dec: -41 36 21.90 (-41.60608d) Equinox: J2000	Proper Motion RA: -11.558 mas/yr Proper Motion Dec: -2.010000025620684 mas/yr Parallax: 0.0011079000000000002" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[K stars]				
	(4)	Ref_for_F770W	RA: 14 08 2.1362 (212.0089008d) Dec: -41 05 20.78 (-41.08911d) Equinox: J2000	Proper Motion RA: -13.906 mas/yr Proper Motion Dec: -10.553999914009182 mas/yr Parallax: 0.0017029" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[K stars]				

Proposal 11864 - Observation 2 - Into the red: characterizing the mid-infrared emission of the circumplanetary disk around PDS70 c

Observation	Proposal 11864, Observation 2: PDS70- Roll1										Fri Mar 13 22:03:34 GMT 2026
	Diagnostic Status: Warning Observing Template: MIRI Imaging										
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (PDS70- Roll1 (Obs 2)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	PDS70	RA: 14 08 10.1546 (212.0423108d) Dec: -41 23 52.57 (-41.39794d) Equinox: J2000			Proper Motion RA: -29.69699999999996 mas/yr Proper Motion Dec: -24.04100005151122 mas/yr Parallax: 0.008897500000000001" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[K stars, Protoplanetary disks]										
Template	Subarray SUB64										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	4-Point-Sets				1	1	POINT SOURCE	POSITIVE	DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F560W	FASTR1	10	400	1	Dither 1	4	1600	1497.772	
	2	F770W	FASTR1	6	600	1	Dither 1	4	2400	1429.676	
Special Requirements	Group Observations 1, 2, 3, 4, Non-interruptible Aperture PA Offset 3 from 2 by 10 to 14 Degrees (Same offsets in V3)										

Proposal 11864 - Observation 3 - Into the red: characterizing the mid-infrared emission of the circumplanetary disk around PDS70 c

Observation	Proposal 11864, Observation 3: PDS70 - Roll2										Fri Mar 13 22:03:34 GMT 2026
	Diagnostic Status: Warning Observing Template: MIRI Imaging										
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (PDS70 - Roll2 (Obs 3)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	PDS70	RA: 14 08 10.1546 (212.0423108d) Dec: -41 23 52.57 (-41.39794d) Equinox: J2000			Proper Motion RA: -29.69699999999996 mas/yr Proper Motion Dec: -24.04100005151122 mas/yr Parallax: 0.008897500000000001" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[K stars, Protoplanetary disks]										
Template	Subarray SUB64										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	4-Point-Sets				1	1	POINT SOURCE	POSITIVE	DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F560W	FASTR1	10	400	1	Dither 1	4	1600	1497.772	
	2	F770W	FASTR1	6	600	1	Dither 1	4	2400	1429.676	
Special Requirements	Group Observations 1, 2, 3, 4, Non-interruptible Aperture PA Offset 3 from 2 by 10 to 14 Degrees (Same offsets in V3)										

Proposal 11864 - Observation 1 - Into the red: characterizing the mid-infrared emission of the circumplanetary disk around PDS70 c

Observation	Proposal 11864, Observation 1: Reference F560W										Fri Mar 13 22:03:34 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(3)	Ref_for_F560W	RA: 14 07 25.5626 (211.8565108d) Dec: -41 36 21.90 (-41.60608d) Equinox: J2000			Proper Motion RA: -11.558 mas/yr Proper Motion Dec: -2.010000025620684 mas/yr Parallax: 0.0011079000000000002" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[K stars]										
Template	Subarray										
	SUB64										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	4-Point-Sets				1	1	POINT SOURCE	POSITIVE	DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F560W	FASTR1	10	400	1	Dither 1	4	1600	1497.772	
	2	F770W	FASTR1	7	600	1	Dither 1	4	2400	1633.964	
Special Requirements	Group Observations 1, 2, 3, 4, Non-interruptible										

Proposal 11864 - Observation 4 - Into the red: characterizing the mid-infrared emission of the circumplanetary disk around PDS70 c

Observation	Proposal 11864, Observation 4: Reference F770W										Fri Mar 13 22:03:34 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(4)	Ref_for_F770W	RA: 14 08 2.1362 (212.0089008d)			Proper Motion RA: -13.906 mas/yr					
			Dec: -41 05 20.78 (-41.08911d)			Proper Motion Dec: -10.553999914009182 mas/yr					
			Equinox: J2000			Parallax: 0.0017029"					
						Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.										
	Category=Star										
	Description=[K stars]										
Template	Subarray										
	SUB64										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	4-Point-Sets				1	1	POINT SOURCE	POSITIVE	DEFAULT	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F560W	FASTR1	8	400	1	Dither 1	4	1600	1225.388	
	2	F770W	FASTR1	6	600	1	Dither 1	4	2400	1429.676	
Special Requirements	Group Observations 1, 2, 3, 4, Non-interruptible										